Discover Education

Review

A review of university chatbots for student support: FAQs and beyond

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Abstract
Within the university sector, student recruitment and enrolment are key strategies as institutions strive to attract, retain and engage students. This strategy is underpinned by the provision of services, applications and technologies that facilitate lecturing and support staff. Universities that offer online learning have a particular incentive to use modern electronic channels such as chatbots to combine high levels of service and availability. This paper details a thorough literature review which investigates recent chatbot implementation developed within a university setting to assist with student queries not related to classroom learning. It initially describes chatbot examples related to student administration activities and teaching and learning, before describing in detail the recent evolution of Natural Language Processing and the more immediate arrival of state-of-the-art generative techniques. The use of chatbot development platforms is also considered and eight university chatbot implementations are compared and considered in detail. Two of the main findings of the review are that datasets associated with closed domains are still difficult to obtain and curate, and that universities unlike other sectors, have not yet realised the potential for chatbot implementation. Finally, the review suggests approaches that take advantage of already existing data and newer generative models and frameworks that can help in the development of student focused chatbots.

1 Introduction

The concept of a conversational agent, commonly referred to as a chatbot, has a history which stretches back to the Turing Test in the 1950's. Since the earliest known implementation of a chatbot called Eliza [1] the general evolution of Artificial Intelligence (AI) continued up to the end of the 1990's. However, with the increasing speed of hardware and the availability of huge amounts of data in the 2000's, these technologies have evolved and developed into sophisticated, modern Deep Learning algorithms. These advances have enabled modern Natural Language Processing (NLP) techniques comprising the understanding of intent from the user through Natural Language Understanding (NLU) and the generation of a response to the user using Natural Language Generation (NLG) as described by Maroengsit et al. [2].

The general advance of NLP has been realised in services such as Apple Siri, Google Assistant and Amazon Alexa which the consumer has become familiar with using. Similarly, where previously email and voice communication would have been the norm, self-service channels such as chatbots are now acceptable if consumers choose to have queries answered in a fast and convenient manner [3].

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This acceptance for chatbots has increased dramatically since late 2022 with the introduction of ChatGPT[1] by OpenAI[2] The Large Language Model or LLM on which ChatGPT is based, has had the effect of both mainstreaming and disrupting the very concept of how Artificial Generative Intelligence (AGI) might be used in all aspects of our lives.

While a review by Baidoo-Anu and Ansah [4] found mixed feelings among educators with regard to ChatGPT for teaching and learning, an earlier assessment by Hussain [5] outlined three broad categories for the use of campus chatbots. These included improvements to teaching, learning and assessment, enhancement of existing information, and reducing friction around day to day activities.

Cunningham-Nelson et al. [6] note the "range of potential benefits, including personalisation and 24/7 instantaneous availability" that chatbots could lend to being applied within the educational sector. However, most of the existing chatbot studies and implementation appear predominantly in sectors such as telecoms, finance, and e-commerce.

Yang and Evans [7] suggest that the absence of implementation might be related to the diverse nature of the educational domain. This view correlates with the description by Hussain [5] who highlights the diverse service categorisations where chatbots could be effectively used. Smutny and Schreiberova [8] also acknowledge that while chatbots have a long history of use as pedagogical agents in educational settings, difficulties were experienced in the specific search for Facebook based educational chatbots. This was attributed to the fact no central service exists to find education specific implementations.

The main aim of this review is to investigate chatbots that are currently being used or have been developed within the closed domain of universities to assist with student queries not related to classroom learning. Many studies exist which deal specifically with the use of chatbots for teaching, learning and assessment but there appears to be no specific systematic literature reviews with regard to their usage outside of the classroom. This review investigates and synthesizes findings from existing university implementations and identifies gaps and opportunities for future development.

Academic and administrative chatbots are discussed in Scope of the Research. Under Current State of Language Models section, several systems including retrieval and generative based models are selected for study. In the Development Platforms section, commercial products relevant to the university sector and how they are contributing to this space are reviewed and commented on. Under the Implementation comparison section, results are discussed in the context of size of the training data used and potential difficulties around the sourcing of the data. Finally, under the Discussion section, a summary of findings is presented. In Conclusions and Future Perspective, the review closes with thoughts on suggested further research into generative techniques as these will allow for the potential re-use of content currently existing within university websites, knowledge bases, and chat forums.

2 Scope of the research

As universities strive to attract, retain and engage students there is also a need to provide applications, services and technologies that facilitate the lecturing and support staff. These service categories are neatly encapsulated by Hussain [5] into three broad categories that could be enhanced with chatbot implementation:

- Enhance how information, advice and guidance can be offered to prospective and existing students.
- Reduce friction for day-to-day activities on campus.
- Provide Opportunities to enhance Teaching, Learning and Assessment.

These enhancements are of particular interest to universities that are increasingly driven by competition in the sector coupled with higher expectation from the student. The student experience, whether that be campus based or online, must be optimised in the face of diminishing budgets, scaling of services and best use of staff [9]. Using the categories identified by Hussain [5] existing chatbot adoption in universities can be considered under the following categories:

- Admissions/Student Recruitment/Administration
- Teaching and Learning

[1] https://chat.openai.com/.
[2] https://openai.com.

2.1 Methodology

This review investigates recent chatbot implementation developed within a university setting to assist with student queries not related to classroom learning. With this in mind, initial searches were carried out on Google Scholar for peer-reviewed conference papers and journal publications relating to the use of chatbots in various industries. This was subsequently narrowed down for papers relating to chatbot usage in the university sector using search terms "chatbot + university" and "chatbot + education". While these initial searches did reveal some interesting work, it was decided that only papers from 2015 onwards could be considered, this date coinciding with the initial release of platforms and tools such as TensorFlow [10] and PyTorch. Papers were then selected on the basis of having being implemented in a university setting outside of teaching and learning. For searches associated with more recent developments in Generative AI, search terms included "university + chatGPT", "closed domain + chatbot + gpt", "chatbot + RAG + university" and "chatbot + langchain + university". All of these searches were focused on finding actual implementation. A final review of these papers determined which were suitable for a more detailed investigation. Crucially, this final review and selection process also sought to establish that implementations for further study were not related specifically to classroom learning.

2.2 Admissions/student recruitment/administration

Universities put great emphasis on the student recruitment process that ensures a constant student pipeline. As a result, Admissions departments are one of the richer areas for the implementation of chatbots in this sector. The student recruitment process may come under strain during the key admission time of late summer, often referred to as 'summer melt' [11]. Choque-Díaz et al. [12] outline the importance of the admissions period for prospective students and the need for focus on student experience by detailing the operation of an online channel offering response to students within 48 hours. Chandra and Suyanto [13] detail the technical development of a WhatsApp based chatbot using frequently asked questions (FAQ) from an admissions department, while Santoso et al. [14] discuss the implementation of an FAQ Chatbot for university admission services. These implementations highlight the fact that FAQs are a particularly effective way to manage the student recruitment process as the same queries often occur repeatedly.

All FAQs are composed of a question and an answer. This suggests one question being asked by the user and the chatbot immediately returning a relevant answer. This is known as a single-turn interaction, with the chatbot producing a response based on an immediately preceding question. A multi-turn interaction enables a suitable response to be generated from the context of multiple previous questions. Though single turn chatbots are highly effective by themselves for independent, recurrent queries in closed domains, Aujogue and Aussem [15] propose a model that enables multi-turn response conversation. In a university chatbot context this might be the user asking a question such as "Can I study mechatronics online?" and then ask, "How much does it cost?". A chatbot would be able to understand that "it" in this case refers to the previous question and could react correctly by displaying the fees for mechatronics.

Apart from admissions and student recruitment, academic administration retains a huge amount of knowledge relevant to students that is managed in various locations such as websites, internal knowledge bases and by individual staff members. Khin and Soe [16] leveraged these resources to implement a chatbot solution for general campus queries including those relating to university location and academic standing, history, conferences, faculties and staff, library, research, and alumni. Similarly, Rana [17] focuses on the existing student population by using existing FAQs along with structured and unstructured information to provide assistance where staff are not available.

2.3 Teaching and learning

Singh [18] states that teaching and learning can be defined as learning, enhanced student engagement, feedback loops, teaching assistants, help and student support and usage analytics. With specific regard to teaching assistants, Satow [19] introduced a model containing six levels for learning facilitation that could be provided by artificial intelligence (AI) based teaching assistants. This progressed from a simple Level 1 of welcoming new learners with personalised messages, through to a Level 6 where the student could receive individual coaching. Yang and Evans [7] highlight that the use of chatbots in education might be limited because the needs and requirements within education may be more diverse than from other domains. This diversity of requirement is demonstrated by those author's research which includes a chatbot simulation game for an online master's programme support and training for a library recommendation system and a helpdesk support system. Similarly, Cunningham-Nelson et al. [6] present an implementation of an FAQ chatbot

which can answer questions associated with assessment, exam requirements and class times. However Nwankwo [20] identifies that the systems and models in place for "academic advising support services"—regardless of the technologies utilised—has been focused on course registration and selection whereas these should relate more to learning outcomes, abilities, and career profiles.

With the implementation of a chatbot called Jill Watson, Goel and Polepeddi [21] dealt with the issues surrounding the support of a Massively Open Online Course (MOOC) and the ability for a chatbot to support many students. Using this implementation, the authors report that the MOOC can respond autonomously to student introductions, post weekly announcements and can answer routine, regularly asked questions.

Lee et al. [22] designed a chatbot called Infobot to support students taking an introductory networking course in a university in Hong Kong. This chatbot is supported on multiple social platforms such as Telegram and Facebook Messenger and allows students to ask questions on course materials, class schedule and class venues.

The work outlined by Cunningham-Nelson et al. [6], Goel and Polepeddi [21] and Lee et al. [22] gives a brief insight into types of implementation within the field of teaching and learning. A systematic review on the use of chatbots in education by Okonkwo and Ade-Ibikola [23] also mentions administration, assessment, advisory and research and development. Of the reviewed articles that were part of this study, 66% related to teaching and learning through the delivery of course content, student engagement, question answering and individualised help.

3 Current state of language models

The previous section outlined chatbot usage scenarios that are common within a university setting, without any specific reference to the design or the type of Language model used. These scenarios are referred to as closed domain and the implementations can perform defined tasks or respond to queries that are relevant and related to that domain. An open domain chatbot—one that has the ability to converse on any topic—is currently more likely to engage the user in a more human-like conversation, so making it suitable for the Turing test [24] as a conversational chatbot. A closed domain chatbot on the other hand may only perform tasks or respond to queries relating to a specific industry, sector, or domain. Accordingly, this review is specifically concerned with the implementation of closed domain chatbots.

The models discussed in this section provide a detailed background to the types that have been used for chatbot implementation in the recent past as well as new, evolving models that are now considered state of the art (SOTA). This review acknowledges the difficulty in discovering university implementations for these models. However, implementation of chatbots in other industries can be found with examples using Artificial Intelligence Modelling Language or AIML [25], Sequence to Sequence or Seq2Seq [26], and Bidirectional Encoder Representations from Transformer or BERT [27, 28]. These examples illustrate diverse usage including customer support for e-commerce, insurance, and finance, as well as a patient consultation chatbot for ophthalmologic diseases.

Examples of open domain, conversational implementations of Generative Pre-Trained Transformer or GPT are readily available and a number of closed domain examples within the university sector have also been found [29, 30, 31, 32].

3.1 Rule based chatbots

Rule based or pattern based chatbots have been the traditional approach to the building of chatbots. The first consideration with the design of any rule-based chatbot will be the capability that it can offer. These are the intents or the broad family of questions that a user might ask. These might be FAQs for a prospective online student at a university that includes groupings under headings such as registration, fees, timetables, exams, and the like. Under each of these headings, a number of intents could be deduced from the user. For example, under "registration" specific intents such as "When do I need to register" or "What documents do I need to register" or "Do I need to be on campus to register" could be identified. Each of these intents would have a specific response and of course each intent could be asked in a number of different ways, known as a pattern; "What documents do I need to register" could also be stated as "Do I need documents to register" or "Do I need my passport for registration."

A rule based chatbot delivers answers to the user by processing a hierarchal set of rules. A chatbot in this case is only as clever as the rule base it contains but is effective in a closed domain [33]. The more rules, the more advanced answers the chatbot can attempt to answer. If the rules do not exist, then the chatbot cannot provide a response. This indicates one of the main drawbacks of a rule-based system; to increase the functionality, all the intents and responses and the various patterns or way in which a question may be asked must be correlated and managed.

```
<?xml version="1.0" encoding="UTF-8"?>
<aiml version="2.0">
  <category>
     <pattern> Hello</pattern>
     <template>Hi, How can I help?</template>
  </category>
<aiml>
```

Fig. 1 Snippet of some simple AIML code structure

3.1.1 AIML

AIML is an open-source XML based, chatbot scripting framework for specifying chatbot content in the form of intents/patterns and responses. This AIML content contains standard and extensible tags that can be processed by an AIML interpreter [34]. Figure 1 below shows a snippet of AIML code including category, pattern, and template tags.

The category is the smallest unit of knowledge in AIML and this always contains input from the user in the form of the < pattern > tag and a response to the user in the form of the < template > tag. As AIML does not allow more than one synonym per category, the example shows that for the variations of the salutation "Hello," a < category > tag would need to be included and expanded with patterns such as "Hi," "How are you" and "Hey."

Currently at version 2.0, AIML has been in existence since the early 2000's and its development is supported through numerous open-source interpreters and libraries and also via online service providers such as Pandorabots.[3] Claiming over one million users, Pandorabots is one on the longest serving AIML chatbot platforms. This may explain the few examples of AIML based university chatbots that have been available for this review. As a rule based framework, AIML appears well suited to FAQs and related solutions. With the implementation of a university chatbot, Khin and Soe [16] built a knowledge base in AIML that included common questions asked frequently by lecturers, students, and their parents. This knowledge base consisted of 970 questions and answers ranged over nine topics. Implementation was through the Pandorabots service.

Ranoliya et al. [35] also outline the development of an AIML based chatbot for university related FAQs and these cover a wide-ranging selection of questions that a chatbot user might ask. Meanwhile Nwankwo [20] ambitiously proposes an AIML based solution for the delivery of an academic advisory chatbot called Adviserbot.

3.2 AI based Chatbots

An AI based chatbot (often also referred to as deep-learning or neural network based chatbot) is a multi-layer perceptron, which uses a relevant text-based training dataset and NLP, to synthesize a response to a user's query using deep machine learning [36]. Unlike a rule-based chatbot where the answer to a question can only be answered if it already exists in the rules designed, an AI chatbot is based on a machine learning model which can be trained to suggest an appropriate answer. The most appropriate model to be used depends on the task to be performed and the data that is available to be trained for the task.

AI chatbots themselves can be broken down into two categories—retrieval based chatbots and generative chatbots. Both categories can use a neural network-based approach [33]. Retrieval based chatbots predict responses based on the existence of question-and-answer pair data that has been derived from another source. A training dataset will provide examples of typical questions and an answer to that question. While the system can be improved through machine learning, by ensuring that variations of questions can be added and the system can be retrained, new answers cannot be automatically generated. Retrieval chatbots are well supported in online platforms such as Microsoft QnA Maker and Google Dialogflow.

[3] https://www.pandorabots.com/.

Generative chatbots are self-supervised models, which use deep learning to create responses that do not already exist in a training dataset. Many of these models (which are described in future sections) such as BERT, GPT, Gemini, Claude and LLaMA are considered SOTA. This status has been realised because of increased computing power [2] and processing techniques and through the use of large datasets available on the internet [37]. Regardless of the model type, all AI models use some common NLU pre-processing techniques to understand the intent of the question being asked by the user.

Text Pre-processing Text Pre-processing is a key step in model building that makes the performance of learning algorithms more efficient. The pre-processing steps ensure the data used by the model is as consistent and as well shaped as possible for further processing. The techniques outlined or variations of the techniques are typically used to deal with the input or the query from the user. Figure 2 gives an overview of common tasks involved in the pre-processing stage using lyrics from a song [38].

Tokenization and all the steps in the pre-processing routine are available in python-based libraries and platforms like spaCy [39], NLTK [40] and TensorFlow. This initial step breaks up or tokenises a text sequence into each individual word, punctuation and non-character. Each token produced is then easily accessible to further steps in the routine. Regularisation normally involves the standardisation of an initial token set so that the content is now stripped of non-characters and punctuation and making all content lowercase. According to Welbers et al. [41] the process of tokenisation and regularisation is often done in one step.

Stop word removal [42] removes certain commonly used words and symbols from text before it is processed by a natural language algorithm. The dictionary of stop words may be refined for question answering as words such as "when" or "what," while stop words, do form a crucial part of intent identification in a user query.

From work initially carried out by Porter [43] and eventually refined into the Porter2 stemmer algorithm [44], stemming is the removal of prefixes and suffixes from words so that what remains is the simple form of the word, the stem. This reduction makes the processing much more efficient. For example—washed, washing reduce to wash, and giving and given reduce to give.

Lemmatisation can be the last step in grouping together different inflected forms of a word so that the word can be seen as the same item known as a lemma (the root word). Unlike stemming, which is only concerned with word reduction, Jivani [45] notes that lemmatisation considers part of speech (POS), and the context of the word in the sentence before producing the lemma. However, this efficiency does come at more of a cost than stemming as lemmatisation requires extra computation for completing word resolution. Examples would be 'mice' to mouse, 'feet' to 'foot', 'sung/sang' to sing and 'is/are' to 'be.'

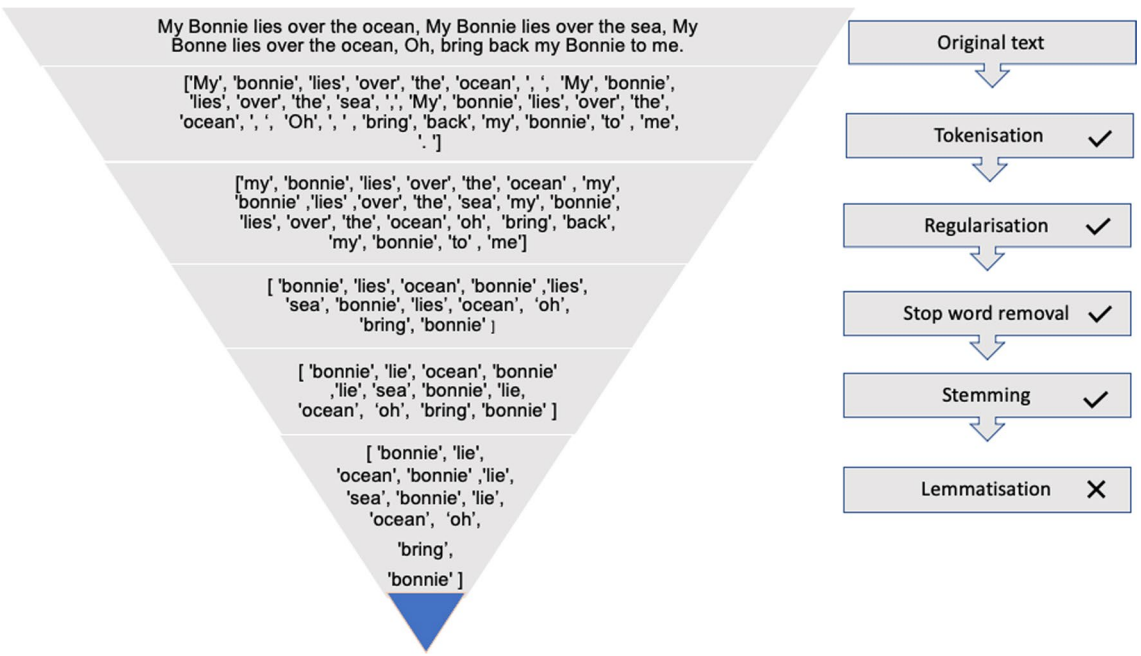

Fig. 2 Typical pre-processing steps

Text Processing Text can be processed by relevant NLU algorithms when it has been optimised as part of the pre-processing steps. The methods below represent examples of what is currently used from a processing perspective, where the actual method used can depend on the problem domain.

Domain classification or topic labelling [46] is used as part of the processing stage if the conversation with the chatbot has the potential to cover multiple domains. This is normally associated with an open domain where the subjects that might be asked might include common conversational topics such as news, weather, or traffic.

Intent Classification is used to understand the meaning of the user's question. The most common form of intent found in any chatbot would be a greeting ("Hello,""Hi,""Hey") and a farewell ("Bye,""Goodbye,""Cheers")—showing that intent can be expressed by the user in different ways. In the context of a university chatbot, typical intents might include registration, fees, induction, timetable, and library. Intent can use algorithms such as bag of words (BoW) or Term Frequency-inverse document frequency commonly known as TF-IDF [47]. Both of these are examples of word embedding techniques, where embedding is the mechanism by which characters, words or sequences of words can be represented mathematically in the form of a vector to capture meaning and relationships. BoW is a statistical language model that extracts features from text by creating a vocabulary of the words that exist in the text and by counting the occurrence of individual words. The location of the words in the text is immaterial. The use of BoW is complemented by TF-IDF which is a weighting that measures the importance of a word in a corpus of text. The weighting comprises two terms; Term Frequency (TF) is the number of times a word occurs in text divided by the total number of words in the document. Inverse Document Frequency (IDF) measures how important a word is by considering that certain terms might occur a lot but have little importance. This is calculated by getting the logarithm of the total number of documents divided by the number of documents with the word occurring in it. The TF-IDF is then calculated by multiplying TF and IDF.

Entity recognition allows words or phrases in the text to be identified for their semantic similarity, word analogy and word similarity. Examples would be the ability to recognise people, names, organisations, and locations in a corpus of text. This process of recognition is enabled by word embedding techniques such as Word2vec and Global Vectors for Word Representation (GloVe). Though similar in outcome, both techniques differ in the way they are trained.

Word2vec in the simplest form is a feed forward neural network model that takes an unlabelled text corpus and produces word vectors as output [48]. The model is trained by using a window size (e.g., five words before and five words after) of nearby words and outputting a single vector for every word in the corpus. This training from nearby words uses two models for prediction—continuous bag of words (CBOW) and the Skip-gram model. CBOW uses the context of the nearby words to predict a target word while Skip-gram uses the target word to predict the context of the nearby words.

GloVe [49] is also an unsupervised model but unlike Word2vec it is trained using a matrix factorisation technique. As it considers the semantic proximity of words (co-occurrence) using the entire corpus it must first construct a large matrix of context for each word of the corpus. This matrix is then factorized so that each row of the matrix now represents a vector for each corresponding word.

Word2Vec and GloVe are examples of context free or static embedding techniques. While both consider the occurrence and location of word in a text, neither takes into consideration the context of the word in the text which ensures that each word will have the same representation irrelevant of the context. Taking the song lyric "My Bonny lies over the ocean" [38], the same representation of the word "lies" means that it could mean Bonny resides over the ocean or that Bonny tells untruths over the ocean or that Bonny is laying prostrate over the ocean.

To overcome this problem of ensuring that words can have different context or meaning and therefore need separate representation, the concept of long-term dependency is introduced in a method known as long short-term term memory (LSTM). This allows context to be derived from characters or words further back in the sequence [50].

LSTMs can remember information by using a mechanism called long term memory or cell state. This cell state comprises a chain like structure comprising four interacting neural network layers which allow the model to load and store information from events which have happened previously. Information flows through this cell state and an activation function predicts a probability in the range 0–1. The information is specifically altered by three sigmoid neural net layers which can allow modification and update of the cell state. Figure 3 shows the structure of a standard LSTM with its key cell state structures included.

As data is processed within the cell, the LSTM can decide to retain information or discard information using gates (the sigmoid and tanh activation functions) including a forget gate, an input gate, and an output gate. The forget gate decides what should be kept from previous steps; the input gate decides what is relevant from the current cell state and the output gate works out what the next hidden state should be. For example, in the sentence "The student completed *her* exam" the subject's gender would be retained in the cell state from a previous sentence or paragraph so that the correct pronoun could be used.

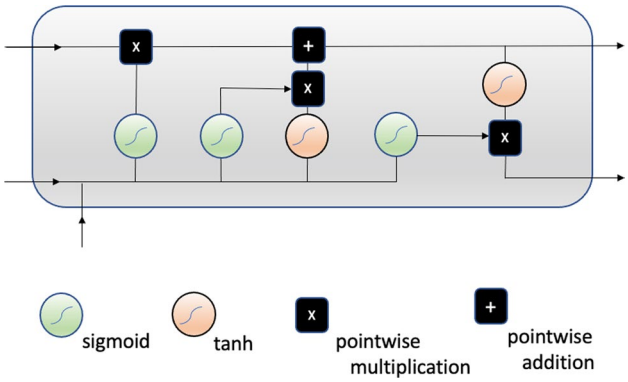

Fig. 3 LSTM showing how a cell state retains or discards information using gates

3.2.1 Retrieval based chatbots

With a better understanding of the basic elements of pre-processing and processing, it is possible to look in closer detail at models using these techniques. Retrieval based chatbots provide pre-defined answers to questions asked by the user. These can be classified in the realm of supervised learning (with labelled data) where with a known input (X) and a known output (Y), an algorithm can be trained to learn the mapping function that predicts the output from the given input. With a new value of X, a value for Y can be predicted by approximating the mapping function. If the accuracy of prediction is high (e.g., greater than 75%), then a user can be assured of a pre-defined answer which satisfies the request. If the level of prediction is low then the chatbot may respond with "I cannot understand your question" or "Please rephrase your question". Refinement of the system may be carried out so questions that fail to produce a relevant answer can be manually associated to an existing answer. Alternatively, a new answer to this type of question can be generated and the system can be retrained.

Datasets for use in a retrieval chatbot may already exist in a question-and-answer format in web FAQs and internal knowledge bases, as well as information generated from backend systems using an Application Programming Interface (API). The dataset goes through the pre-processing and then processing, identifying the intent of the question and the important parts of the sentence using entity recognition techniques such as word2Vec or GloVe. With the model trained, a question posed to the system will generate a response that has the greatest probability of being correct.

As part of a solution for assisting new students during the admissions process Mislevics et al. [51] developed a retrieval-based system called Leo that allowed a knowledge base of FAQs to be created. Alias et al. [52] used a pattern matching technique for an Academic chatbot called Acabot for a unique retrieval chatbot implementation. Rana [17] explores the concept of a hybrid chatbot which Marthur and Lopez [53] describe as using retrieval and generative approaches to respond to user input if there is no match in the former. It is also worth noting that many chatbot development platforms (discussed in a future section) primarily use retrieval-based methods which allow for rapid development of solutions. While implementation might be fast, it does mean that maintenance is required for the upkeep and monitoring of the FAQs. The predefined question and answer pairs need to be maintained and added to for new queries which are posed by users.

3.2.2 Generative based chatbots

Whilst retrieval chatbots can only provide pre-defined answers, generative chatbots use self-supervised deep learning to create responses that may not already exist in the training data**.** While many variations exist for the implementation of a generative chatbot, the core process is similar, with a flexible response generated that is based on the processing of very large amount of unlabelled training data. This data originates in web-based information such as websites, books, computer code and social media and allows a language model to be trained or have a pre-training phase. This pre-training phase might use a dataset like the Cornell Movie Dialog Corpus [37] consisting of conversations from over 600 movie scripts. An effective dataset for models that might be fine-tuned—the ability to improve on a particular task or domain—as a question-answering model is SQuAD, a collection of question–answer pairs derived from Wikipedia articles [54]. These large datasets, using often a combination of supervised learning with labelled data and self-supervised learning (SSL) with unlabelled data, form an integral part of the training for many generative models. The following sections review generative techniques, some of which are currently producing SOTA results.

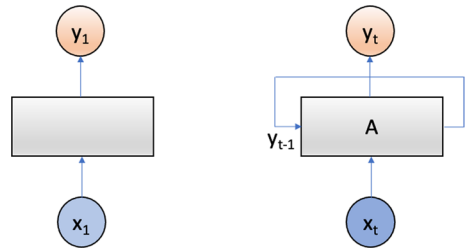

Fig. 4 Traditional fully connected RNN on left compared to rolled up RNN on right

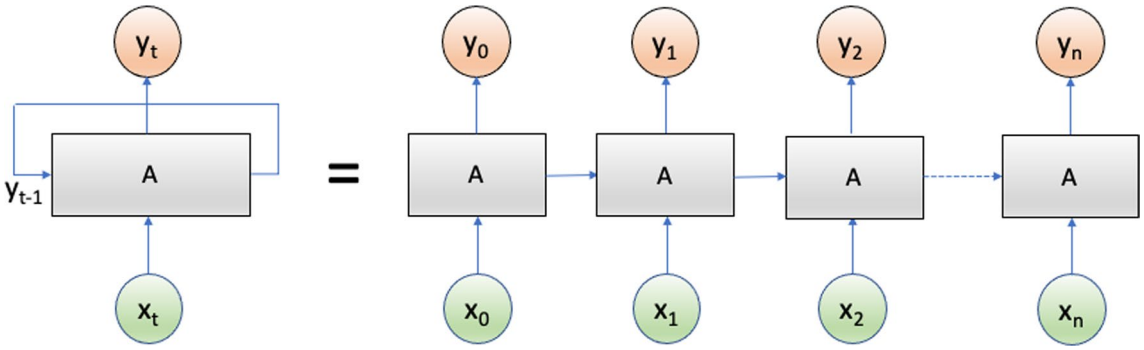

Fig. 5 An unrolled RNN

Frequent Pattern (FP) Growth Frequent Pattern (FP) Growth algorithm is an unsupervised model which is commonly used for discovering patterns or relationships in large amount of data. Popular within the data mining community, FP Growth uses a divide and conquer approach to represent data items and the associations between data items in a tree form. Before the tree is constructed, data items which occur below a minimum are removed. Each node of the tree then represents a data item and how it is associated with other data items. More frequent occurring items having an increased chance of sharing the same data. Once constructed, the tree can be mined recursively to get the frequent pattern and Pattern growth is returned by concatenating these frequent patterns from condition FP tree [55]. Alias et al. [52] trained an Academic chatbot called Acabot using a corpus of data derived from the social media interaction between students and an internship coordinator. Using a variation of an FP-Growth model called Frequent Adjacent Sequential Pattern or FASP [56], the authors compared the results against an N-gram model where they found encouraging evidence to suggest that a user's intent could be extracted even with a limited vector size.

Sequence to Sequence (Seq2Seq) Seq2Seq is a supervised learning model that is traditionally used to convert an entire input sequence of words into an output sequence of words. The model is derived from a popular neural network called a Recurrent Neural Network (RNN). RNNs are particularly useful in working with contextual or sequential data. It takes output from a previous timeframe, which is allowed to be persisted and fed as input to the next frame in the network.

Shown on the left in Fig. 4 is a traditional fully connected RNN network with an input (x_1) and output (y_1). On the right is a rolled-up timeframe with input (x_t) and outputting a value (y_t) where *t* is timeframe and *A* is an RNN. The loop (y_{t-1}) takes the output of the previous timeframe and feeds this back as input into the next frame in the network.

Figure 5 shows the network unrolled. Output from a certain layer can be saved and this can then be used as input to the next immediate layer so that it can help predict the output from the current layer.

In terms of NLP and RNN, the input sequence is a sentence from which each word is pre-processed and processed. The output sequence is a group of predicted words which are turned into a sentence [57]. This ability of an RNN to handle sequential data and to be able to consider input from current as well as from immediate previous input makes it ideal for NLP related tasks.

A standard RNN implementation, however, will have trouble in training with a gradient based learning method, a very popular optimisation algorithm used in deep learning [58]. The weights on hidden layers can become increasingly small meaning that weights do not change value and as a result, further training of the model is curtailed. This Vanishing Gradient issue was highlighted by Hochreiter [59]. Another issue with a standard RNN implementation is that a layer only receives input from a short number of its previous neighbours so dependency is short term. This is fine if prediction is required for the final word in a short sentence such as "the cat drinks *milk*"—the distance between the word to be predicted and its previous neighbours is short.

The Seq2Seq LSTM model is also particularly useful where the inputs and outputs have different lengths. Figure 6 shows a standard RNN containing a single hidden layer. RNN A is shown as a repeating module which is taking previous hidden state and the current input to form a vector. The repeating modules show tanh being used as an activation function.

A standard RNN only has a single hidden layer, whilst Fig. 3 shows the structure of a repeating module in a standard LSTM. Another improvement can potentially be made by employing an Attention mechanism in the form of a bidirectional LSTM or BiLSTM [60]. This consists of two LSTMS—one of which takes input and processes in the normal forward direction and another which processes in a backward direction. This enables the model to read an input context in both forward and backward directions so that the algorithm can retain context of words that follow and precede words in a sentence. This ability of the Attention mechanism to manage and quantify word interdependence can mitigate with the issue of a long sequence of words in a sentence.

Palasundram et al. [61] primarily focus on the research and analysis of optimal settings for a Seq2Seq model for natural answer generation from university FAQs. The authors report that Seq2Seq models have not been used very much for the development of educational chatbots and that they find little in the way of experimentation and analysis of how Seq2Seq models deal with the variation of input in question generation, for example regarding of synonyms. Khin and Soe [62] implement a university chatbot using Seq2Seq but in this case with Attention mechanism. While the authors state that the model accuracy is not very high, it is found that the result using Bilingual Evaluation Understudy or BLEU [63]—a measurement used for evaluating the quality of the text output of a chatbot—with Attention mechanism, is better than just using a baseline Seq2Seq implementation. Chandra and Suyanto [13] implement a chatbot for university admissions also using Seq2Seq and with Attention mechanism. As with Khin and Soe [62], the experiments showed improved BLEU results using Attention mechanism. The authors additionally found that reversing the input sentence also led to a small improvement in BLEU results.

Bidirectional Encoder Representations from Transformers (BERT) RNN based algorithms process words in a sequential fashion from left to right and from right to left, but this process involves multiple steps for making decisions about words which are far apart from each other. In contrast to this sequential determination, the Transformer model [64] developed by Google, applies a self-attention layer within each step of a small constant number of steps. This self-attention layer computes or relates the relationship between each word in a sentence, regardless of the position. Each word is compared to every other word in the sentence, the result of which is an attention score for each word [65].

The Bidirectional Encoder Representations from Transformers (BERT) model is based on the Transformer architecture and is a deeply bidirectional unsupervised learning model that can be pre-trained. This ability to pre-train allows the fine-tuning of models with one additional output layer. The pre-training process does not use left-to-right or right-to-left language models but two unsupervised tasks including Masked Language Model (MLM) and Next Sentence Prediction (NSP). The bidirectional nature of MLM allows the entire sequence of words to be read at once, with each word able to indirectly "see itself" by randomly masking a percentage of the input tokens and then predicting the value of these masked tokens. This masking of input tokens is the mark of an autoencoding transformer. NSP supports the understanding of the relationship between sentences by allowing pre-training on labelled sentences A and B where 50% of the time B is a sentence that follows A and the other 50% of the time B is a random sentence that does not follow A [66]. For a pre-trained model, BERT typically uses versions of SQuAD. This dataset consists of 100 k + questions that have been crowd worked so that questions correspond to a piece or span of text or alternatively that no answer is available [49]. Rana [17], pursues an ambitious academic chatbot implementation called Eaglebot that uses several data sources from the Georgia Southern University website and is composed of different models in a hybrid implementation. The author

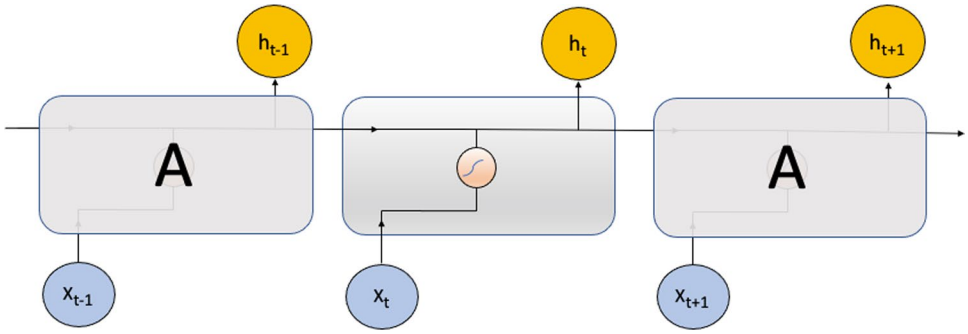

Fig. 6 Standard RNN with single hidden layer and repeating modules, where tanh is the activation function

establishes the best performance of models in different use cases including structured tabular data, questions and answers, and unstructured textual data.

Generative Pre-trained Transformer (GPT) Whereas Seq2Seq models take a sequence and output another sequence, Generative Pre-trained Transformer or GPT, introduced by OpenAI in 2018, is an autoregressive LLM which can predict future values based on past values. It is the first language model to have the potential to perform AGI with the ability to learn how to perform a specific task even though the base model may not know of that task. The model uses semi-supervised learning in a two-stage training framework comprising self-supervised pre-training and supervised fine-tuning. Since GPT enables a wide range of uses including text summarisation, translation and question answering, the authors wished to keep the model free of task specific customisation and instead employ an approach where structured inputs such as questions and answers are converted into a standard ordered sequence that the model can process [67]. GPT-4, released in March 2023 is the fourth generation of this language model and outperforms both previous LLMS and most SOTA systems [68]. GPT-4 reputedly uses 1.7 trillion parameters, far surpassing the first iteration of the initial model in 2018 which had 110 million parameters [69]. It also uses the same underlying language model as used in previous versions but is now also able to understand and process sources other than text such as images. With the introduction of ChatGPT in November 2022, OpenAI enabled conversational interaction via a chatbot with the GPT model allowing the user to converse, ask questions and be assisted with creative tasks such as writing. This has created a new awareness within society of how AGI, and LLMs specifically, will be used for all aspects of modern life.

At the same time there has been considerable interest in the literature as to the influence ChatGPT with regard to teaching and learning. A review by Baidoo-Anu and Ansah [4] on the topic, discovered fixed feelings among educators. These feelings may relate on the one hand to benefits such as interactive learning and assessment preparation but on the other to negative aspects such as bias and the generation of incorrect, nonsensical, or unreal information, also known as hallucination. These negative aspects of LLMs are described by Bender et al. [70] who suggest that these issues may be remedied by the careful curation of datasets relevant to the task at hand.

Though the literature is rich with sources of findings on use of ChatGPT with regard to teaching and learning, it has been largely absent of work associated with implementation of GPT or ChatGPT in a university setting outside of teaching and learning. Bieletzke [29] discusses a prototype implementation of GPT-based solution whose initial functionality allows the student to generate thesis titles. They also suggest how it could be used for giving potential students advice on the application process, costs and only answering questions relevant to a specific programme. They outline how this could be achieved by mediating access to GPT through the OpenAI API via a custom front-end, and through data provided by relevant university Content Management Systems (CMS).

Lappalainen and Narayanann [71] describe the building a chatbot for a University Library, using GPT models via the OpenAI API. This solution consisted of data scraped from existing web-based library resources along with a manually created dataset of FAQs containing 2,200 rows of information. The authors also used Langchain,[4] a development framework for harnessing the use of LLMs in applications.

Finally, a number of works identify the use of GPT along with Falcon [72] in conjunction with RAG, a technique introduced by Lewis et al. [73]. This allows information from external sources such as databases, documents, or websites to be retrieved by an LLM in response to a query, rather than actually fine-tuning the model with this information directly. Cherumanal et al. [30], Odede and Frommholz [31], and Siragusa and Pirrone [32] describe the use of RAG for building chatbots that answers queries relating to specific university programmes.

Gemini, Claude, and LLaMA Gemini is the name of a family of LLM models, formerly known as BARD, first introduced by Google in February 2023. While Gemini [74] like GPT, uses cross-modal reasoning which uses data from different modalities, it also claims to advance the SOTA of benchmarks commonly used for LLMs.

The launch of Claude by Anthropic in March 2023, saw the concept of Constitutional AI being used to train large models. According to the authors [75] this enables LLMs to be both helpful and harmless. The family of Claude models achieves this by using supervised learning and reinforcement.

Finally, the introduction of LLaMA (Large Language Model Meta AI) by Meta in July 2023 saw the introduction of a series of LLM's that the authors claim was competitive with the results from GPT-3 on most benchmarks. Touvron et al. [76] outline the changes they implemented in these LLaMA models which included attaining SOTA performance using openly publicly available data, a smaller number of parameters (7, 13, 33 and 65 billion) and making the models available to the Open-Source community.

[4] https://www.langchain.com/

3.3 Model overview

Table 1 shows a high-level breakdown of closed domain university chatbots models that have been available to be examined as part of this review. These have been chosen as they are example of chatbots that assist with student queries not related to classroom learning. The data used in these models and the results obtained are discussed in a later section.

As can be seen in Table 1, just one of the models is rule based. Even though AIML is a proven technology for rules-based solutions and is suitable for the delivery of FAQ based chatbots it may be the case that the technology is less relevant with the advent of more modern approaches. It is not surprising that three of the models are encoder/decoder based as this underpins generative methods. Of the models—Seq2Seq—might be viewed as the link between retrieval and generative approaches. FAQ based retrieval chatbots may still continue to be a pragmatic option for solution delivery. Similarly, a Seq2Seq approach has utilised FAQ data to illustrate the potential that generative implementations will continue to achieve. The release of BERT and GPT has introduced the concept of pre-training. These models have already been trained using massive datasets and corresponding computing resources. The corresponding fine-tuning can then be completed on a potentially smaller question and answer dataset that is relevant to the closed domain.

Table 2 shows a breakdown of the comparative advantages and pitfalls of the models reviewed, within the context of chatbot implementation.

3.4 Development platforms

The platforms described here provide an opportunity for disparate development and research of chatbots in any sector, as well as solutions catering specifically to the university market. Cunningham-Nelson et al. [6] give a high-level comparison using generalised parameters of common chatbot development platforms including QnA Maker, Dialogflow, RASA, Wit.ai, and Botkit.ai. Some of these platforms may initially appear as non-developer focused tools utilising a retrieval model. However, by allowing easy generation of question/answers, button/menu-based selection opens, single-turn and more complex multi-turn interaction as well allowing the development of bespoke functionality through the availability of an API, they become quite powerful. Microsoft QnA Maker is typically used in conjunction with Microsoft Language Understanding (LUIS) service. QnA Maker enables the creation of question answer pairs while LUIS enables intent to be gathered from the user through a Transformer based language model called Turing Universal Language Representation Model (T-UVR). Like QnA Maker, Google provides DialogFlow as a completely hosted chatbot development environment, using BERT for NLU.

While it is difficult to ascertain what specific GPT models are being used by OpenAI in their ChatGPT offering, OpenAI do provide access to the latest GPT models via API access for developers. Different GPT models may be used depending on the domain and the application, and fine-tuning is also available for some models. Platforms like QnA Maker and DialogFlow may serve the non-developer community, but there are a wide variety of options available for developer and researchers that wish to develop solutions programmatically or to develop new models and tweak existing ones. RASA provides a conversational development framework that is available as a paid, hosted service as well as an open-source development platform.

Similarly, Wit.ai provides consumer focused developer functionality with multiple programming language support and API integration. Botkit.ai is purely an open-source development toolkit for building chatbots and providing

Table 1 High-level breakdown of closed domain chatbot models

Name	Type	Pre–training	Fine-tuning	Architecture	Model Type	Training
AIML	Retrieval	N/A	N/A	Rule Based	N/A	N/A
Seq2Seq	Generative/retrieval	×	✓	RNN/LSTM	Encoder/Decoder, Auto-regressive	Supervised
BERT	Generative	✓	✓	Transformer	Encoder/Decoder, Auto-encoding	Self-supervised
GPT	Generative	✓	✓	Transformer	Encoder/Decoder, Auto-regressive	Self-supervised
FP Growth	Retrieval	✓	✓	FASP	N/A	Unsupervised

Table 2 Advantages and pitfalls

Model Name	Advantages	Pitfalls
AIML	• Long history of implementation • Easy to implement • Use of XML for representation	• Data needs to be entered manually or through editing tool • Intent identification based on pattern matching • Superseded by newer techniques
Seq2Seq	• Can be used for retrieval/generative purposes • Can use attention for improved performance	• Can have difficulty with longer sequences of input data, as context may be lost by the end of the sequence
BERT	• Uses pre-training/fine-tuning • Available via opensource	• Requires more training data • Models can be very large which may affect performance
GPT	• Uses pre-training/fine-tuning • Needs less training data • More examples of closed domain implementation becoming available • Integration to data sources using RAG possible	• Only available via paid for API but comparable models available via open source • Potential for generation of incorrect results, bias
FP Growth	• Useful for study of non-traditional approach	• Highly Experimental approach

integrations with popular messaging platforms. Access to SOTA models and large datasets for the researcher, are provided through environments such as Google Colabs[5] and Hugging Face.[6] These environments provide Jupyter notebook access to machine learning libraries such as Google TensorFlow and Facebook Pytorch without the need for having massive local GPU and CPU configurations for model training.

It is interesting to note a number of providers that provide specific white labelled chatbot solutions for the university sector including Mainstay[7] (formerly AdmitHub) and Unibuddy.[8] Mainstay describes itself as an engagement platform that uses behavioural intelligence to provide a broad range of solutions for areas including admissions & enrolment, retention, and college access. Out of the box integration is offered to well-known Learning Management Systems (LMS) such as Canvas[9] and Blackboard[10] as well as developer access via API. Mainstay does show deep integration possibilities with numerous student use-cases including an implementation called Pounce[11] in conjunction with Georgia State University and CSUNny[12] with California State University. Unibuddy focuses entirely on the area of student enrolment and engagement with a live chat platform. The platform has a deep focus on the use of metrics for measurements like conversion rates of applications and other engagement criteria. Out of the box integration is provided with the Slate CRM system.

3.5 Implementation comparison

Having outlined the model types and high-level implementation details for a number of university chatbots, it is now relevant to compare these implementations to those from other closed domains. Of particular interest here is the data used for training purposes, the availability of this data and the methods used to evaluate the results.

Metrics have always been used for evaluating the results from NLP models and these continue to evolve with SOTA LLMs. BLEU by Papineni et al. [63] is a well-established metric in the field of machine translation (MT) where text in one language is converted into an equivalent in another language. It is used in implementations carried out by Palasundram et al. [61], Chandra and Suyanto [13], Khin and Soe [62] and Kapociute-Dzikiene [77].

The F1 Score is used by Nuruzzaman and Hussain [26], and Yu et al. [28] to evaluate their Seq2Seq and BERT implementations respectively. F1 is derived from two summary measures—precision, and recall. Precision describes the proportion of entities that are returned correctly while recall describes the proportion of entities that should be returned that are actually returned [78].

In the work completed by Alias et al. [52], an n-gram comparison—sequences of *n* adjacent words—was evaluated against the results of an FP-growth technique from the authors. With the use of GPT, Cherumanal et al. [30] chose to use ROUGE-1 and BERTScore. Like BLEU, both techniques are used for evaluating text similarity. While ROUGE [79] applies n-gram methods similar to BLEU for text summarisation, BERTScore [80] calculates semantic similarity by using meaning and context as well as word position.

Given the range of evaluation methods highlighted above, it is noted how Maroengsit et al. [2] examined chatbot implementation with regard to broad headings of content, user satisfaction and chat functionality but suggested that the evaluation methods used would be determined by their type and goal. Liu et al. [81] further highlight that while metrics from machine translation and text summarisation have had significant usage, techniques that use embedding-based metrics along with taking into account context, should be considered.

Table 3 outlines and compares key details of the closed domain chatbot models reviewed as part of this study. Eight are from university settings, three are from the financial sector and one is from the medical field.

A common theme with practically all of the university chatbots being reviewed is the small amounts of data that are being utilised, with the difficulty of access to training data being specifically mentioned by many authors. Palusundram et al. [61] manually curated a small experimental dataset containing one hundred pairs of questions and answers. Some of this data was used for training (seen questions), and the remaining data was used to create five categories of unseen questions so that the model results could then be studied:

[5] https://colab.research.google.com/.
[6] https://huggingface.co.
[7] https://mainstay.com/.
[8] https://unibuddy.com/.
[9] https://www.instructure.com/.
[10] https://www.anthology.com/.
[11] https://theuia.org/sites/default/files/2021-07/GSU-Research-Report.pdf.
[12] https://www.csun.edu/admissions-records/csunny.

Table 3 Comparison of educational chatbots with those from other closed domains

Author(s)	Bot Name	Application Domain	Response Generation	Model	Dataset	Evaluation	Evaluation Score	Result
Palasundram et al. [61]	N/A	University	Generative	Seq2Seq	100 pairs	BLEU	0.95	Word based model performs better than character-based model
Khin and Soe [16]	N/A	University	Rule Based	AIML	970 pairs	Human	N/A	Implementation to be refined using ML techniques
Chandra and Suyanto [13]	N/A	University	Generative	Seq2Seq with Attention	2,900 pairs	BLEU	0.4468	Attention improves BLEU score
Alias et al. [52]	Acabot	University	Retrieval	FP-Growth	537 pairs	N-Gram comparison	N/A	Descriptive intent pattern discovery with small-term vector size
Khin and Soe [62]	N/A	University	Generative	Seq2Seq with Attention	5,000 pairs	BLEU	0.41	Attention improves BLEU score
Cherumanal *etal* [30]	Walert	University	Generative	Falcon-7B	106 questions	BERTScore, ROUGE-1	0.771 0.671	Improved performance noted using training variations of questions versus RAG
Siragusa and Pirrone [32]	Unipa-GPT	University	Generative	GPT	506 documents, 269 pairs	Human	N/A	Much experimentation with corpora and question asking techniques
Rana [17]	Eaglebot	University	Generative	BERT	100 pairs 400 pairs	Unknown	0.65 (FAQS) 0.8 (unstructured)	BERT-Base provided most accurate results with answers for FAQs, unstructured passages
Nuruzzaman and Hussain [26]	Intellibot	Insurance	Generative	Seq2Seq	10,115 pairs	F1 score	0.9857	When compared with other chatbots and evaluated for accuracy, Intellibot performed better
Yu et al.[28]	AVA	Finance	Generative	BERT	22,630 pairs	F1 score	0.754	BERT large performed best of models compared
Lee et al. [27]	OCS	Medicine	Generative	BERT	1176 pairs	N/A	N/A	No formal evaluation carried out
Kapociute-Dzikiene [77]	N/A	eBusiness	Generative	BERT	567 pairs	BLEU	0.513	Best results from BERT when compared with other architectures

1. Place the words in different order so that the word order of the questions is different from that of the seen questions.
2. Replace words so that one of the words in the seen question is substituted with another in vocabulary word.
3. Use words with spelling variations so that one of the words in the seen question is spelt differently.
4. Add additional words in question so that an additional in-vocabulary word is added to a seen question.
5. Shorten the question sentence so that one of the words is taken out from a seen question.

These experiments used four variations of the dropout rate along with fixed hyperparameters and BLEU as the metric for evaluation. For what the authors agree is the primary research question "*How well can the recurrent neural network (RNN) based Seq2Seq model handle the various categories of education chatbot?*"—it was noted that the model generated the highest BLEU score of 0.95 with a dropout rate of 40% on a word-based model. If given a carefully curated initial dataset, the model was able to generate synonyms. This was also the only work that investigated and compared word-based embedding to character-based embedding and found that performance was improved with word-based embedding.

Khin and Soe [62] utilised a corpus containing 5000 question answer pairs (in a total of about 10,000 sentences) of an admissions team and Facebook messenger from a university in Myanmar. Data was also manually collected from students and parents that wanted to inquire about the university. Of the resultant pairs, 4700 were used for training and 300 for testing when evaluated using an attention-based model with a batch size of 64. Testing and evaluation discovered that the model performed relatively poorly with a BLEU score of 0.41.

Chandra and Suyanto [13] utilised question and answer data from an Indonesian university admissions department WhatsApp group consisting of 2903 messages to build a Seq2Seq model using attention. The number of actual individual question answer pairs obtained from the result data is not disclosed but the conversations are broken into 2506 pieces of training data and 397 piece of testing data. Each conversation contained one or more question and answer pairs. In the final comparison of BLEU scores, the attention based model showed improved results but still with a relatively low score of 0.4468. The authors suggest that this score may be related to the fact that Malay is a syllable-rich language as opposed to English which uses character n-grams as sub words.

Alias et al. [52] used data from WhatsApp conversations relating to student internship queries to populate their model named Acabot at a Malaysian university. This unsupervised model consisted of twelve conversations comprising 537 conversation lines. In comparing the unsupervised Frequent Pattern Growth (FP-Growth) algorithm against an N-gram model, the authors found encouraging evidence to suggest that a user's intent could be extracted even with such a limited vector size.

Cherumanal et al. [30] developed Walert as a prototype conversational agent for the answering of FAQs at an Australian university. Two techniques were evaluated and compared, with the Falcon 7B model used for both of the approaches. For FAQs and the intent-based approach, specific instructions (known as prompt engineering) were used to generate semantic variations of one hundred and six questions, and these were used to train the model. A RAG approach was then used to evaluate how successfully answers could be generated from passages of text contained in a knowledge base. The authors established that the intent-based approach for FAQs with known questions performed better using BERTScore and ROUGE-1 (0.77.1 and 0.671 respectively) but that the RAG approach was likely to generate better answers to inferred questions.

Siragusa and Pirrone [32] utilised GPT in their implementation of Unipa-GPT, a chatbot to assist prospective students to find out more specific information about programme offerings in an Italian University. The LangChain based solution used qualitative evaluation to compare the results returned from the fine-tuning of a GPT 3.5 model to the same model using RAG. The authors report that the fine-tuned model did not return significant improvements on the RAG model. However, they did comment that a prompt that uses the previous conversation along with the next question performed well. They also mentioned a very relevant hallucination for university chatbots where the system was asked for a bachelor course but returned the details of a master's course by the same name.

Rana [17] posed the question of how a chatbot could fulfil requests across the entire domain of a university, utilising Dialogflow's NLU toolkit to understand user intention, BERT and other methods for FAQs and finally BERT and BiLSTM for unstructured passages. For FAQs, over 400 question answer pairs were extracted from existing web based materials with questions being compared to queries compared using TFIDF and semantic similarity techniques. For unstructured passages, a test dataset comprising 100 + questions was used to evaluate three BERT models and a BiLSTM model. The author showed that the BERT-base model displayed improved results over the other BERT candidates and baseline BiLSTM model giving an accuracy of 65%. However, this improvement came with the overhead of having to break a document into sentences and paragraphs manually.

Whereas university chatbot implementation had access to small question–answer datasets, large amounts of data seemed to be more accessible for chatbot development in the finance domain. With their implementation of a Seq2Seq dialogue-based chatbot for the insurance industry called Intellibot, Nuruzzaman and Hussain [26] had access to use a 10k question answer pair dataset which was custom built. When compared and validated with other chatbots from the literature, Intellibot provided an F1 score of 0.9857.

With a chatbot called AVA, Yu et al. [28] used a BERT implementation for financial services with access to over 22 k question answer pairs. This dataset was collected, curated and generated over the course of twelve months and provided an F1 score of 0.754.

While implementations in the financial field are expected, it is interesting to note a medical implementation for making patients more aware of ophthalmological diseases, in this case macular degeneration. Lee et al. [27] describe the development of a BERT based model with the intention of reducing the burden on doctors and nurses for recurring questions. A Korean version of SQuAD (KorQuAD) was used to train the model and 1,100 question answer pairs were developed in conjunction with a medical specialist. While the authors state that the system shows promising results and could be used by patients, formal evaluation has yet to be carried out. This patient-focused implementation may be a good example of where human interpretation of results may be seen to be more informative than a formal scoring system [82].

An e-commerce focused chatbot implementation was presented by Kapociute-Dzikiene [77] to provide details on a company's products and services. Comparing BERT results with different encoded-decoder architectures including Seq2Seq, difficulty was admitted in having only 567 question answer pairs in their dataset. Even though a BLEU score of only 0.513 was achieved, the authors state that the results are good enough to be applied in practice.

Having looked at a number of these implementations, some general observations can be made. There appears to be a sharp divide between the quantity and availability of question answer pairs between university and some other closed domain chatbots. This could be for a number of reasons. Large quantities of relevant information might not exist in a curatable format (e.g., email) in the educational domain. Costs may be prohibitive if curation of a dataset is required—it is probably unfair to compare the budget of an admissions department of a mid-range university with a financial services company capable of sustaining a curation project over twelve months [28]. Finally, it is interesting to note a high correlation from many of the papers reviewed with the desire to actively deploy the solution for user benefit.

4 Discussion

The major focus of this review is to investigate recent chatbot implementation developed within a university setting to assist with student queries not related to classroom learning.

Eight university chatbots, using a number of different model approaches, were judged suitable for review and for an understanding to be gained of their implementation. Just three of these used LLMs and techniques that would now be considered to be SOTA. This is a small number when compared with other searches on domains including customer service, finance, and travel. These numbers may indicate that universities are employing turn-key solutions for chatbot development. This is evidenced by the existence of companies such as Unibuddy and Mainstay who have a specific focus in the education sector.

4.1 Small datasets

One of the main findings from the review is the small amount of training data in evidence across implementations. In only one of the implementations did the test data size exceed 5,000 pairs of question answers—and this was for a specific AIML rule-based solution. The difficulty of having access to training data is noted by Ezen-Can [83], Sumikawa et al. [84], Kapocuite-Dzikiene [85] and Sumikawa et al. [84] while Adamopoulou [36] specifically notes the difficulty in finding suitable training data for closed domain chatbots. However, it is reassuring to see that the solutions with small datasets reviewed here that used transformer based generative solutions still considered results to be acceptable.

4.2 Absence of working generative solutions in university sector

As mentioned previously, there are several companies providing turnkey question answering solutions in the university sector. One could conclude that conventional question answering or retrieval chatbots might still be the norm in the sector despite the advancements of generative based solutions. This may be influenced by the fact that some models

are still novel and models like GPT are only available via a paid API. There also may be potential difficulties in access to relevant knowledge bases and data. Finally, universities may be worried about implementing GPT based solutions due to potential language biases previously identified in other sectors [86] as well as broader impacts relating to the potential inability of the reader to distinguish it from human-written text [87].

4.3 Use of sections and paragraphs of text with BERT implementations

Among the implementations reviewed was work on a hybrid BERT implementation by Rana [17]. The author suggests that rather than full pages of text it might be more appropriate to feed in paragraphs or sections to help with scalability and speed of retrieval of responses. This is corroborated by Yang et al. [7], who suggest that paragraph retrieval outperforms article retrieval by a large margin as articles contain many non-relevant sentences that serve as distractors.

4.4 Non-technical rationale for model selection

Of the implementations reviewed only Alias et al. [52] suggested the creation of a new original approach for a model because of the high costs associated with the utilisation of a chatbot platform from companies such as Microsoft or Google. However, this implementation does not appear to be a usable chatbot for university but a specific study of intent classification using models are more normally found in general data science.

It has already been noted that the literature provides many results on the use of ChatGPT with regard to teaching and learning but relatively few that relate to administrative use within a university setting. However, recent work by Bieletzke [29], Cherumanal et al. [30], Odede and Frommholz [31], and Siragusa and Pirrone [32] is encouraging for the authors. It shows that SOTA work is being carried out utilising prompt engineering, fine tuning versus RAG comparison, and development frameworks for LLMs such as LangChain for the development of student focused systems outside of teaching and learning.

5 Limitations

The precise focus of this review may have introduced a number of limitations to the work. It has already been noted that while the literature is rich with studies relating to the pedagogy associated with ChatGPT for teaching and learning, there is very little relating to admissions, student recruitment and administration. It was also identified early in the review that along with a number of companies providing bespoke chatbot solutions for areas such as student recruitment, platforms also exist that can potentially provide functionality for non-developers. While Yang and Evans [7] had suggested this might be the case, the focus of this review on implementation on admissions, student recruitment and administration may have further narrowed the field of study. Finally, some common technical limitations were identified in the eight implementations reviewed with regard to the quantity and availability of data sets in education when compared to other closed domains such as those found in Finance.

6 Conclusions and future perspective

This represents a review of SOTA chatbots implementation in a university setting to assist with student queries not related to classroom learning, with a specific focus on the model architectures and datasets used. Using the broad categories described by Hussain [5], the author focused on areas of implementation such as Admissions, Student recruitment, Administration along with Teaching and Learning. Of these areas, Admissions and Student recruitment appear to be a relatively underserved category in terms of chatbot development.

While including an overview of traditional rule-based approaches and AIML, there was a detailed exploration of AI based chatbots including retrieval and generative methods. A brief comparison of the advantages and pitfalls associated with each method was outlined and it was noted that retrieval methods for FAQs will continue to have a strong presence in closed domain implementation. Continuously evolving and novel generative techniques that have advanced since the launch of ChatGPT will be need to be assessed and evaluated for suitability for deployment.

As previously mentioned, eight university chatbots were chosen for review based on the type of model used and area of implementation. The fact that Yang and Evans [7] suggested that use of chatbots in education might be limited because of the diversity of needs and requirement in education was found to be the case. One of the factors mentioned for this was the ready availability of development platforms that enable fast implementation for non-developers but with the potential requirement of regular maintenance. However, it is exciting to note that more recent implementations from Cherumanal et al. [30] and Siragusa and Pirrone [32], showcase the use of SOTA LLMs along with initial comparisons of the results of fine-tuning models and RAG.

As well as a limited number of suitable implementations for review, it has been found that the size and availability of suitable datasets within universities seems to be problematic when compared to other sectors. However, Adamopoulou [36] suggests this as a problem that may be faced with any closed domain implementation.

Based on this review and findings there are some interesting approaches that could be taken with the development of a university chatbot, for example for admissions and student recruitment. Regardless of the model approach, the availability of data which offers the potential to be curated into useful datasets is crucial. This might include website-based course information, online Questions and Answers for prospective students, email interactions and CRM system-based information. These could form the basis for relevant, curated datasets but with the unknown factors of cost and time of generation.

A hybrid approach using retrieval and generative techniques can offer a pragmatic solution. Standard FAQs could be dealt with via retrieval while other textual content could be targeted for generative techniques. A novel approach would be required for the latter. Course information containing discrete data elements such as duration, cost, and subjects to be studied each semester would need to be formatted as content suitable as training data.

While the authors have considered relevant chatbot implementation over the past five years, they are conscious that ChatGPT has had considerable influence since launching in November 2022. Up to 2019, generative models derived from BERT have been a valid starting point for experimentation. Derivations such as RoBERTa [88], DistilBERT [89] or XLNet [90] could also have been considered. SBERT by Reimers and Gurevch [91] has also offered intriguing possibilities for the derivation of semantically meaningful sentence embeddings from both FAQs and other textual content.

Finally, it is also worth noting the importance of the development frameworks and techniques associated with LLMs that continue to evolve at a very fast rate. This includes LangChain with its support for RAG and synthetic data generation, as well as prompt engineering techniques. Ensemble techniques [92]—the process of combining multiple models to achieve better results, and frameworks for orchestrating complex AI tasks such as AutoGen[13] and CrewAI[14]—while not mentioned specifically in the literature for university chatbots may also be worthy of further exploration and investigation.

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[13] https://microsoft.github.io/autogen/.
[14] https://www.crewai.com/.

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